Students Assesment Number_____

THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MOSHI MUNICIPAL COUNCIL
SECONDARY EDUCATION EXAMINATIONS SYNDICATE
032 CHEMISTRY
FORM TWO MOCK ASSESMENT

Tuesday 28th June 2022 Am **Time 2:30Hours**

Instructions

1. This paper Consist of sections A, B and C with total of 10 questions
2. Answer all questions in both sections
3. All writings should be in blue/black ink
4. All diagrams should be drawn in pencil
5. Write your assessment number at the top right corner

SECTION A (15Marks)

1. For each of the items (i)-(x), choose the correct answer from among the given alternative and write its letter in the box provide.

i. The welders prefers to use non luminous flame for their work simply because

(A)It is available (B) it is easy to transport
(C) produce very hot flame (D) can be made by kerosene

ii. Spatula in the laboratory is used for scoping what types of substances?

(A)Liquid and gases (B) solids and liquids
(C) Powdery and gases (D) Solids and powdery

iii. There are two particles inside the nucleus which one contributes the net change of that nucleus?

(A)Dalton (B)Protons (C) Electrons (D) Neutrons

iv. Why oxygen as one of the components of air Is unique?

(A)It has ability to burn (B) it support combustion
(C) it is diatonic gas (D) combine with carbon dioxide

v. Why it is necessary to boil drinking water?

(A) To make it tasteless (B) To remove impurities
(C) to kill micro-organism (D) To make it taste less

vi. How do chemists refer to a mixture of milk and water?

(A)Miscible solution (B) Suspension (C) Immiscible solution (D) Emulsion

vii. How can one prevent rusting in fragile instruments like camera?

(A)By galvanization (B) By using silica gel (C)By using oil (D) By using ethanol

viii. The oxidation state of metallic element is always

(A) Negative (B)Positive (C)Zero (D) Neutral

ix. Which of the following is not man-made product of applying chemistry?

(B) Sugar (B) Fertilizer (C) Milk (D) Vaccines

x. The atomic number of element R is 9 What is its electronic configuration?

(A)2:4:3 (B) 2:5:2 (C) 2: 7 (D) 2:6:1

ANSWERS

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. (a) Match each item in List A with a correct response in List B by writing the letter of the correct response below the corresponding item number in the table provided

LIST A	LIST B
i. Spirit in water	A. Winnowing
ii. Sodium Chloride in water	B. Layer separation
iii. Oil in sunflower	C. Fractional distillation
iv. Ammonium Chloride crystals in sand	D. Picking
v. Maize and husk	E. Solvent extraction
	F. Sublimation
	G. Decantation
	H. Evaporation

LIST A	i	Ii	iii	iv	v
LIST B					

SECTION B: (70MARKS)

3. (a) Mrs. Msema kweli was addressing the villagers about use of water for economic values. What 4 values will she tell the villagers

i. _____

ii. _____

iii. _____

iv. _____

(b) Is an air a mixture or a compound? Give 4 reasons

i)_____

_____

ii)_____

_____

iii)_____

_____

iv)_____

_____

4. (a) Do your think there is any reasons for providing first Aid to the Victim? If yes give four reasons.

i._____

_____

ii._____

_____

iii._____

_____

iv._____

_____

(b)State the uses of the following items in first aid Kit.

i. The pair of scissors

_____

_____

ii. The whistle _____

iii. Petroleum Jelly _____

iv.Sterile gauze_____

5. (a)State three sub- atomic particles and their respective charge.

i._____

ii._____

iii. _____

(b)State the four modification of Dalton`s atomic theory

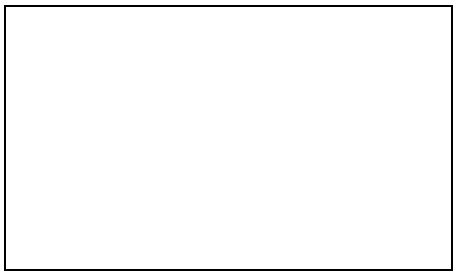
i. _____

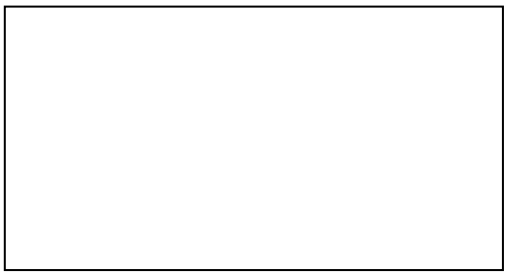
ii._____

iii. _____

iv. _____

(c)Draw and write the electronic configuration for the atoms having the following atomic numbers.

(i) $_4$Be 

(ii)$_{14}$Si 

6. (a)Distinguish the following substances.

i. Homogenous mixture from heterogeneous mixture.

_____

_____

_____

_____

ii. Miscible from immiscible liquids.

_____

_____

_____

_____

iii. Saturated from unsaturated solution.

_____

_____

(b)Give two chemical tests for pure water

i. _____

_____

ii.

_____

_____

7. (a)Asha want to select a good fuel for domestic uses in her home what things should she considered? (Mention six (6)

i. _____

ii. _____

iii. _____

iv. _____

v. _____

vi. _____

(b)Classify the following fuel either primary or secondary fuel.

(i) Coke_____

(ii)Crude oil_____
(iii)Kerosene_____
(iv)Natural gas _____

8. (a)What properties of hydrogen gas that made it to be used in the following?
   i. As a fuel _____
   ii. To fill weather balloons_____
   iii. Manufacture of hydrochloric acid _____
   iv. Manufacture of margarine _____

(b)) Environmental pollution in most of rural areas in Tanzania is caused by using charcoal and firewood as a fuel. What will be the two alternative source of energy, they supposed to use for environmental conservation? (give reason for each)
i._____
ii_____
(c) Why fossil fuels are referred to as non-renewable energy resources? (Give two reasons)
i. _____
ii. _____

9. (a)You are provided with the following atoms, for each fill the informations below:-
(i) $^{35.5}Cl$
protons = _____
Neutrons=_____
Electrons=_____
(ii) ^{1}H
protons = _____
Neutrons= _____
Electrons=_____

(b) Define the following
i.Atom_____
_____
ii. Energy
_____
_____
iii.Atomicnumber_____
v. Isotopes_____
_____

SECTION C (15 MARKS)

10a)i. What do you understand by the word "Relative atomic mass"?

_____

_____

ii). Carbon has main isotopes, $^{12}_{6}C$ and $^{14}_{6}C$ with relative abundances 98.89% and1.11% respectively. Calculate the relative atomic mass of Carbon.

_____

_____

_____

_____

_____

_____

b)i. Draw a diagram of laboratory preparation of oxygen gas from hydrogen peroxide and manganese IV oxide

(ii) Write the word equation for the reaction above (a)i

_____

_____

c) Answer the following questions with reference to the first 20 elements of the periodic table

i. Give the chemical symbol of element having:

- The smallest atomic size_____
- The largest atomic size _____

ii.Identify the element which are:

- Metals having three shells of electrons each _____

_____

- Metals having 1 electron in the valence shell

_____

- Noble gases _____